ISYE 6420 Homework 4

Question 1

1a)

Setting up probability density distributions.
Likelihood:

$$L(X, \theta, \mu) = \sqrt{\frac{\theta}{2\pi}} e^{\frac{-\theta(x)^2}{2}}$$

Likelihood at $x = -2$:

$$L(X, \theta, \mu) \propto \theta^{\frac{1}{2}} e^{-2\theta}$$

Prior and proposal distributions are Gamma distributions.
Prior $\sim Ga(1/2, 1)$:

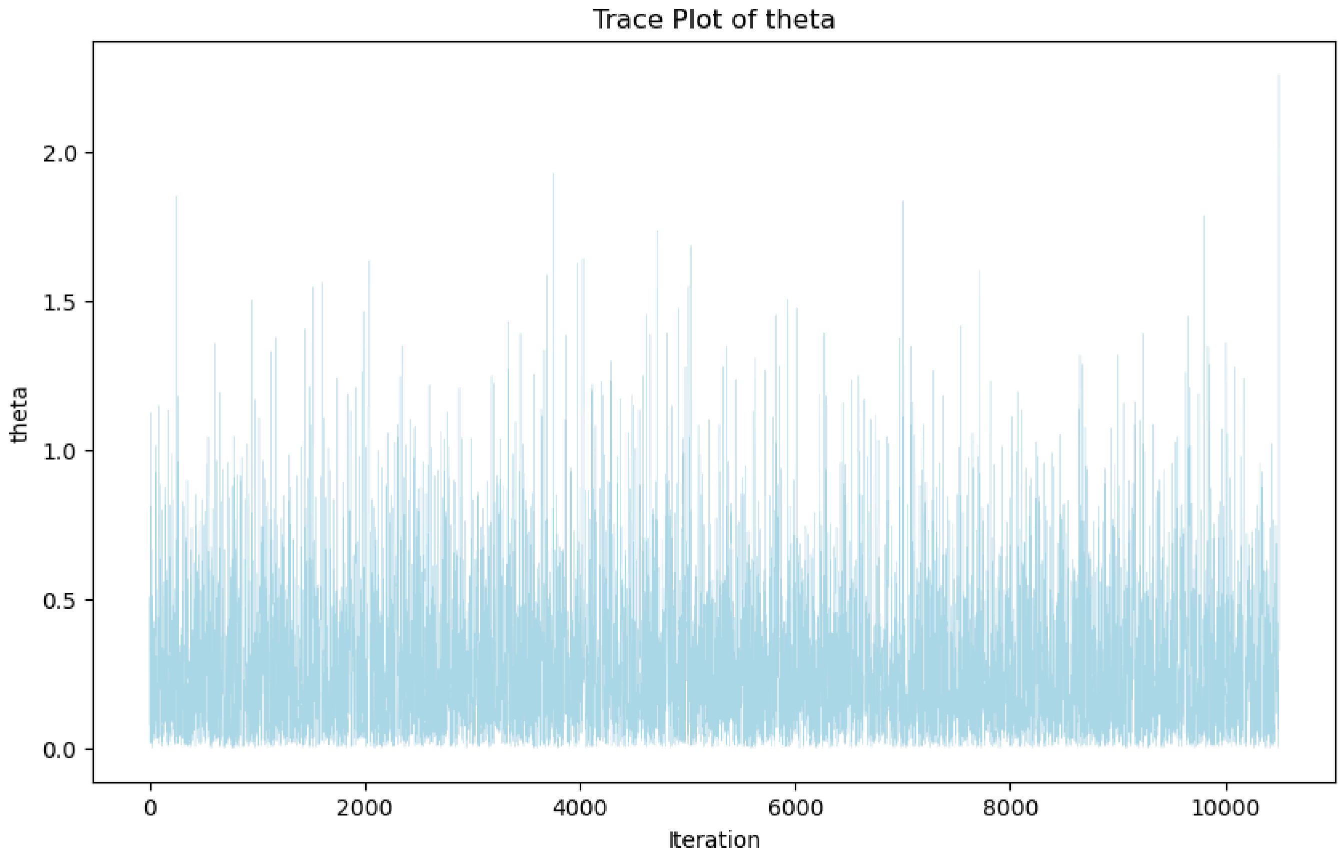
$$\pi(\theta|1/2, 1) = \frac{1^{\frac{1}{2}}}{\Gamma(1/2)} \theta^{-\frac{1}{2}} e^{-\theta} = \frac{1}{\Gamma(1/2)} \theta^{-\frac{1}{2}} e^{-\theta}$$

$$\pi(\theta) \propto \theta^{-\frac{1}{2}} e^{-\theta}$$

Proposal distribution:

$$q(\theta^{'}, \theta) \propto \theta^{'\alpha-1} e^{-\beta\theta^{'}}$$

After testing different a and b values, the initial values of a=1 and b=4 are chosen for now to showcase the Metropolis-Hastings workflow with a suitable acceptance criteria.



1b)

After burning the first 500 samples, the acceptance rate is around 0.86.
In the case of a non exact distribution, take the mean of all our theta samples. In the case of (a=1, b=4), the Bayes estimator is ~ 0.33.
HDI Credible Interval calculated using Arviz package: [0.00, 1.14]

1c)

Finding the exact posterior distribution:

$$\pi(\theta|X) \propto L(\theta|X).\,\pi(\theta)$$

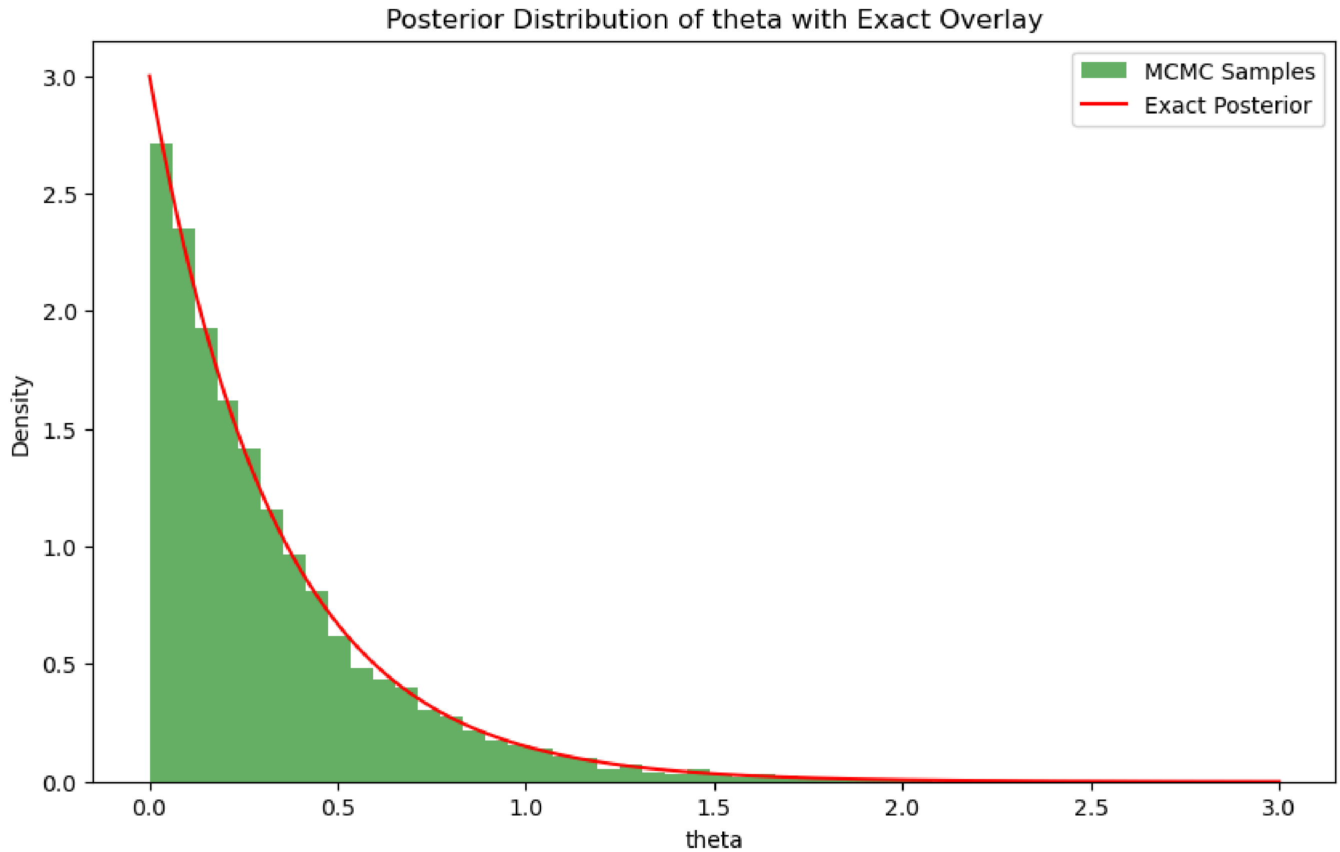
$$\pi(\theta|X) \propto (\theta^{\frac{1}{2}}e^{-2\theta})(\theta^{-\frac{1}{2}}e^{-\theta})$$

Using the kernel method to compare indices:

$$\theta^{\alpha-1}e^{-\beta\theta} \propto (\theta^{\frac{1}{2}}e^{-2\theta})(\theta^{-\frac{1}{2}}e^{-\theta})$$

$$\theta^{\alpha-1}e^{-\beta\theta} \propto e^{-3\theta}$$

$$\alpha = 1, \quad \beta = 3$$



The posterior distribution is a gamma function with a=1, which is equivalent to an exponential distribution.

1d)

Bayes estimator (posterior mean) for Metropolis-Hastings, given it's a exponential distribution and we know the exact distributions (a=1, b=3):

$$\mu_{posterior} = \frac{a}{b} = \frac{1}{3}$$

Acceptance Rate: 1
Bayes Estimator (posterior mean) of theta (after burn-in): 0.33
HDI Credible Interval (using ArviZ) for exact posterior: [0.0, 1.17]
As expected the accepted rate is 100% as this is an exact distribution. For properties of the non-exact posterior distribution tested please see part 1b.

Question 2

2a)

To find the conditional distribution first we need to compute the joint distribution, which would be the likelihood multiplied by the prior for both θ and y_0.

Prior for θ:

$$\pi(\theta) \sim Uniform(0,1)$$

Here the probability density function = 1 for as, pdf = $\frac{1}{b-a} = \frac{1}{1-0}$.
We do not need to consider a prior for y_0 as it is a latent variable and not a parameter.

Joint distribution:

$$f(\theta, y_0, \bar{y}) = p(\bar{y}|\theta).\,\pi(\theta).\,\pi(y_0)$$

$$f(\theta, y_0, \bar{y}) \propto \frac{n!}{y_0!(y_1-y_0)!y_2!y_3!y_4!}(\frac{1}{2})^{y_0}(\frac{\theta}{4})^{y_1-y_0+y_4}(\frac{1-\theta}{4})^{y_2+y_3}$$

Conditional Distribution of y_0

Isolating just the terms with y_0:

$$\pi(y_0|y_1,\theta) = \frac{y_1!}{y_0!(y_1-y_0)!}(\frac{\theta}{4})^{y_1-y_0}(\frac{1}{2})^{y_0}$$

Normalise with y_1 probability: $(\frac{1}{2}+\frac{\theta}{4})^{y_1}$

$$\pi(y_0|\theta,y_1) = \binom{y_1}{y_0}\frac{(\frac{\theta}{4})^{y_1-y_0}(\frac{1}{2})^{y_0}}{(\frac{1}{2}+\frac{\theta}{4})^{y_1}} = \binom{y_1}{y_0}\frac{(\frac{\theta}{4})^{y_1-y_0}(\frac{1}{2})^{y_0}}{(\frac{1}{2}+\frac{\theta}{4})^{y_0}+(\frac{1}{2}+\frac{\theta}{4})^{y_1-y_0}}$$

$$\pi(y_0|\theta) = \binom{y_1}{y_0}(\frac{\frac{\theta}{4}}{\frac{1}{2}+\frac{\theta}{4}})^{y_1-y_0}(\frac{\frac{1}{2}}{\frac{1}{2}+\frac{\theta}{4}})^{y_0}$$

Setting $\phi = \frac{\frac{1}{2}}{\frac{1}{2}+\frac{\theta}{4}}$

$$\pi(y_0|\theta,y_1) = \binom{y_1}{y_0}\phi^{y_0}(1-\phi)^{y_1-y_0}$$

$$\pi(y_0|\theta,y_1) \sim Bin(y_1,\phi)$$

Conditional Distribution of θ

Isolating just the terms with θ:

$$\pi(\theta|y_0,\bar{y}) \propto (\frac{\theta}{4})^{y_1-y_0+y_4}(\frac{1-\theta}{4})^{y_2+y_3}$$

We can bring the $\frac{1}{4}$ term out and it becomes a normalising constant which can be ignored.

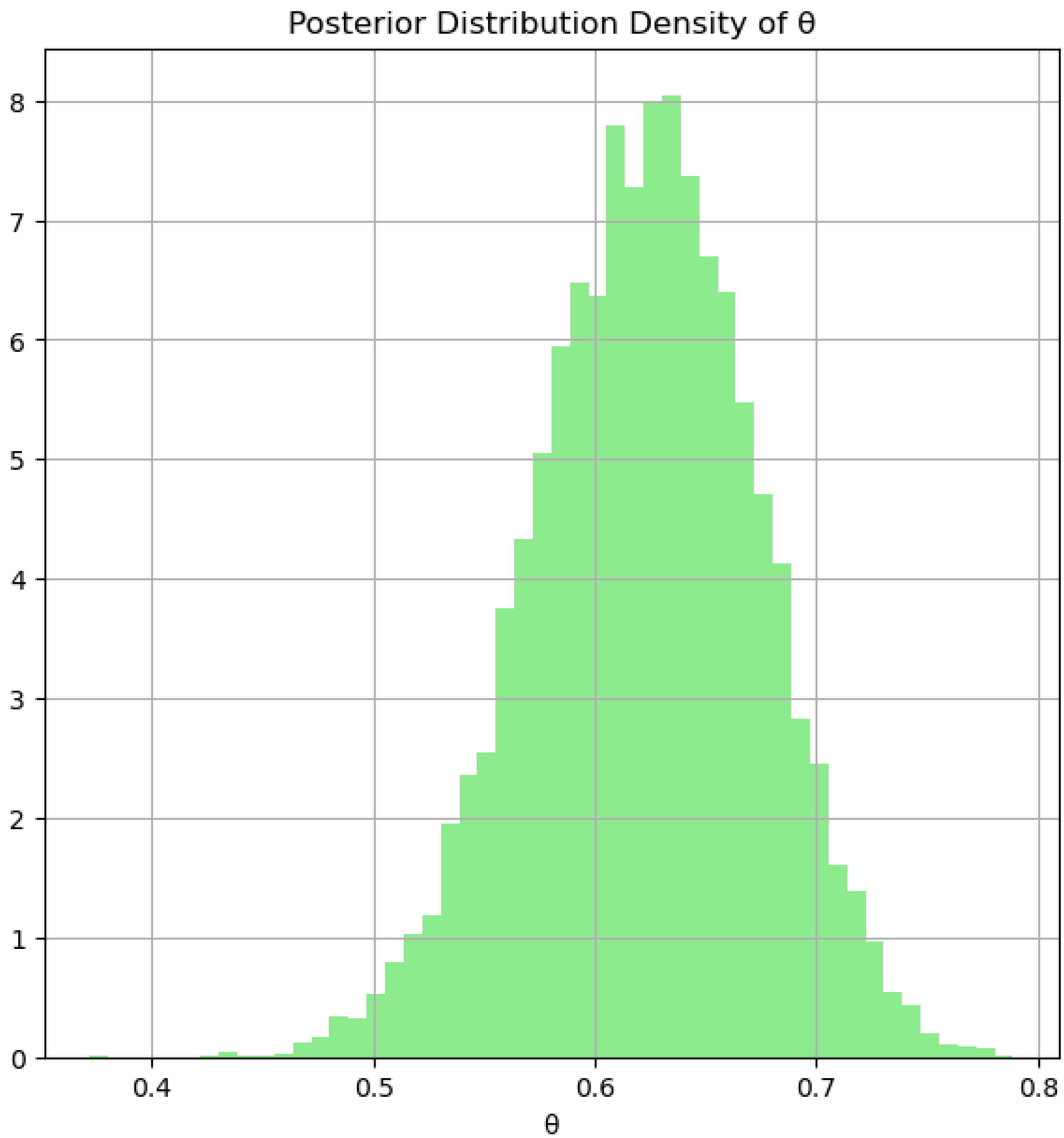
$$\pi(\theta|y_0,\bar{y}) \propto \theta^{y_1-y_0+y_4}(1-\theta)^{y_2+y_3}$$

This resembles a Beta distribution:

$$\pi(\theta|y_0,\bar{y}) \sim Beta((y_1-y_0+y_4+1),(y_2+y_3+1))$$

where $\alpha = (y_1 - y_0 + y_4 + 1), \beta = (y_2 + y_3 + 1)$.

2b)



2c)

Point Estimate (posterior mean) of theta: 0.622
95% equi-tailed credible interval: [0.52, 0.72]
Probability within set: 0.95

Question 3

3a)

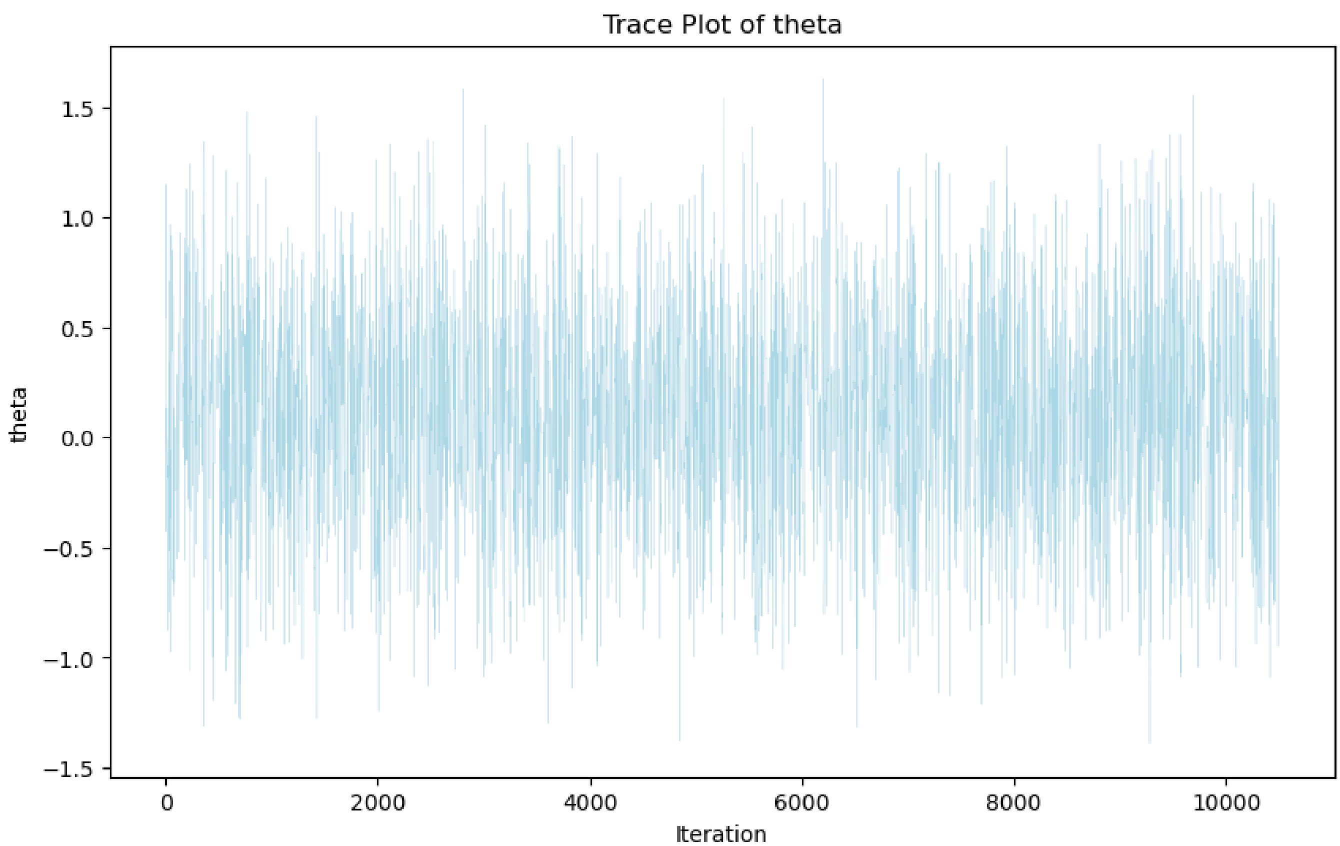
Given the likelihood equation given, we know its distribution is the following:

$$f(y|\theta) \sim N(\theta, \frac{1}{\tau})$$

The Proposal distribution is assumed to be a uniform distribution between -2 and 2. Considering we want a stationary distribution with specific bounds, a uniform distribution is the simplest to implement, although a truncated normal distribution could also offer as an interesting proposal distribution given it meets our criteria.

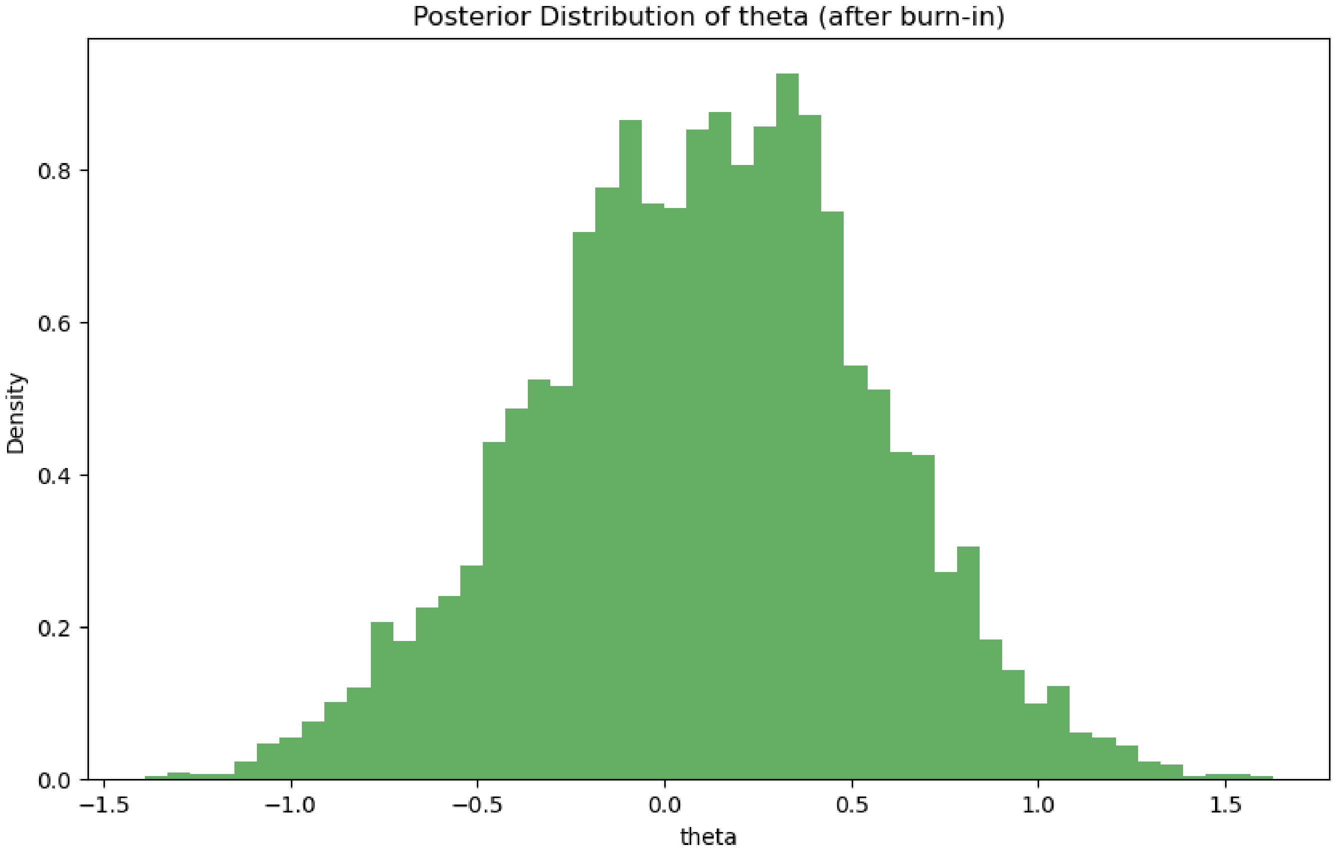
$$q(\theta', \theta) \sim Uniform(-2, 2)$$

3b)



3c)

After burning first 500 samples, the histogram resembles a normal distribution.



3d)

Bayes estimator (posterior mean) of theta (after burn-in): 0.108

95% Equi-tailed Credible Interval: [-0.80, 0.97]

Probability within set: 0.95